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Research Paper

The role of ethical leadership in promoting internal whistleblowing among nurses: A dual-mediation model analysis of psychological safety and reporting attitudes

Ibrahim Abdullatif Ibrahim [a,b], Indrajit Doddanavar [c], Samah Mohamed Abdelrahim [d], Mennat Allah G. Abou Zeid [e,f,*]

[a] Department of Nursing Sciences, College of Applied Medical Sciences, Shaqra University, Shaqra, Saudi Arabia
[b] Department of Nursing Administration, Faculty of nursing, Mansoura University, Egypt
[c] Department of MBA, Jain College of Engineering and Research, Visvesvaraya Technological University, Belagavi, India
[d] Department of Nursing Administration, Faculty of Nursing, Damietta University, Damietta, Egypt
[e] Nursing Administration and Education Department, College of Nursing, Prince Sattam Bin Abdulaziz University, Al-Kharj, Saudi Arabia
[f] Department of Nursing Administration, Faculty of Nursing, Ain Shams University, Cairo, Egypt

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ABSTRACT

Objectives: This study aimed to investigate the dual mediating roles of psychological safety and reporting attitudes in the association between ethics-centered leadership and internal whistleblowing among nurses.

Methods: A multicenter, cross-sectional study design was employed, involving a convenience sample of 273 nurses among five university hospitals from August to October 2024. Data were collected using a demographic questionnaire, Ethical Leadership Scale, Psychological Safety Scale, Positive Attitudes Toward Whistleblowing Scale, and Internal Whistleblowing Scale. Pearson correlation and structural equation model were conducted to test the hypothesized dual mediation model.

Results: The average score of nurses for ethical leadership was 4.14 ± 0.57, psychological safety was 3.80 ± 0.64, reporting attitudes was 4.12 ± 0.67, internal whistleblowing was 3.93 ± 0.52. Ethical leadership had a significant effect on psychological safety ($\beta = 0.38$, $P < 0.001$), reporting attitudes ($\beta = 0.44$, $P < 0.001$), and internal whistleblowing ($\beta = 0.21$, $P = 0.009$). Furthermore, psychological safety ($\beta = 0.26$, $P < 0.001$) and reporting attitudes ($\beta = 0.27$, $P < 0.001$) significantly influenced internal whistleblowing among the nurses. The indirect effect of ethics-centered leadership on internal whistleblowing intermediated by psychological safety was significant ($\beta = 0.10$, $P < 0.001$). Similarly, reporting attitudes demonstrated a significant mediating effect ($\beta = 0.12$, $P = 0.003$).

Conclusions: Ethical leadership encourages internal whistleblowing among nurses by fostering psychological safety and positive reporting attitudes. Therefore, healthcare managers should prioritize ethical leadership development, enhance psychological safety, and foster constructive reporting attitudes that enable nurses to report ethical violations confidently, without apprehension.

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What is known?

- Ethical leadership is essential for promoting internal whistleblowing among nurses.
- Psychological safety serves as a mediating mechanism through which ethical leadership encourages whistleblowing behaviors.

What is new?

- It empirically demonstrates that ethical leadership indirectly enhances whistleblowing behavior by simultaneously improving nurses' sense of psychological safety and positive attitudes toward error reporting.

* Corresponding author. Nursing Administration and Education Department, College of Nursing, Prince Sattam Bin Abdulaziz University, Al-Kharj, Saudi Arabia.
E-mail address: m.abouzeid@psau.edu.sa (M.A.G. Abou Zeid).
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- By incorporating reporting attitudes as a distinct mediator, the study offers novel insights into the cognitive and emotional mechanisms influencing nurses' decisions to report wrongdoing in hierarchical, high power-distance healthcare environments.

1. Introduction

In the healthcare context, upholding ethical standards and promoting organizational integrity are critical for safeguarding patient well-being and optimizing organizational performance. Ethical leadership, characterized by moral integrity, fairness, and the promotion of ethical behavior, plays a pivotal role in preserving patient safety, upholding professional integrity, and fostering a culture of responsibility [1,2]. In the nursing profession, where ethical decision-making is inherent to patient advocacy, ethical leadership is particularly salient in shaping responses to misconduct, including internal whistleblowing [3,4]. Internal whistleblowing refers to the act of reporting ethical violations or misconduct that occurs within an organization, plays a vital role in maintaining organizational integrity, and ensures patient safety [5,6]. Nurses are uniquely positioned to identify and report incidents compromising ethical or professional standards. Internal whistleblowing, when supported, serves as a critical mechanism for promoting transparency, safeguarding patient safety, and improving healthcare quality [7,8]. Despite its importance, internal whistleblowing remains underutilized, often due to fear of retaliation, lack of psychological safety, or negative attitudes toward reporting [9,10].

While existing research has examined the direct correlation between ethical leadership and internal whistleblowing [11]. A paucity of studies elucidating how ethical leadership translates into actual reporting behaviors among nurses has been addressed recently. El-Gazar et al. [12] has made a valuable contribution by highlighting the intermediary role of psychological safety between ethics-centered leadership and whistleblowing intentions. A shared belief held by members of a team that the team is safe for interpersonal risk-taking [13]. However, the study by El-Gazar et al. [12] focuses solely on psychological safety, leaving other potential mediators underexplored, such as reporting attitudes.

Reporting attitudes refer to nurses' evaluative judgments about the personal and organizational consequences of reporting wrongdoing. It reflects their beliefs about whether whistleblowing will lead to constructive change or punitive consequences, their emotional responses to reporting, and their readiness to engage in such behavior. Drawing on Ajzen's Theory of Planned Behavior [14], attitude is conceptualized as a proximal determinant of behavioral intention. It encompasses nurses' perceptions of whether whistleblowing will result in positive organizational change or personal retaliation, emotional responses to potential reporting, and readiness to act. Attitude was selected over other potential mediators because it captures an individual-level, modifiable cognitive construct that bridges the gap between environmental influences (e.g., ethical leadership) and behavioral outcomes (e.g., whistleblowing). Moreover, it provides greater explanatory precision regarding the decision-making process involved in reporting, especially in sociocultural contexts where moral norms and organizational trust vary significantly. The originality of our study lies in its focus on reporting attitude as a mediator, which has been largely overlooked in nursing literature. Reporting attitudes reflect nurses' beliefs about the consequences of whistleblowing and their subjective evaluation of those consequences, making them a critical factor in determining whether nurses report misconduct [15,16].

While psychological safety has been previously established as a mediator, it remains central to this study because of its foundational role in high-stakes healthcare environments. Including psychological safety in our model allows replication of earlier findings across diverse healthcare settings, enhancing robustness and generating context-sensitive insights. In Egyptian hospitals, hierarchical decision-making, centralized authority, and high power-distance cultural norms are well documented and can suppress nurses' willingness to speak up about errors or unethical practices [17,18]. Within this context, psychological safety serves as a critical mediator, fosters an environment in which nurses feel secure to raise concerns or report errors without fear of punishment [12]. Therefore, incorporating psychological safety as a mediating variable in this study is theoretically justified and contextually relevant to the sociocultural realities of Egyptian hospitals.

To advance this line of inquiry, the current study addresses these gaps by introducing a dual-mediation model that examines the parallel roles of psychological safety and positive reporting attitudes in the relationship between ethical managerial practices and nurse whistleblowing. In doing so, it not only replicates the findings of previous studies, such as El-Gazar et al. [12], but also expands the literature by incorporating reporting attitudes, thus offering a more holistic understanding of the pathways through which ethical leadership shapes whistleblowing behavior. This study is theoretically grounded in Bandura's Social Cognitive Theory [19], which emphasizes the triadic interaction among cognitive factors, environmental influences, and behavior. Ethical leadership is an environmental stimulus and social model whose ethical behavior communicates acceptable norms. Nurses internalize these norms through observation, which influences their cognitive appraisals manifested as perceived psychological safety and positive attitudes toward reporting. These cognitions then inform behavior, namely the act of whistleblowing. Thus, the theory directly informs the selection of mediators and the proposed pathways.

By integrating psychological safety and attitudes toward whistleblowing as dual mediators, this study proposes a comprehensive explanatory framework for understanding the mechanisms linking ethical leadership to internal whistleblowing among nurses. This dual-mediation model is a novel contribution that extends beyond single-mediator approaches and enriches the theoretical landscape with context-specific insights from Middle Eastern healthcare systems.

This study aimed to investigate the potential parallel mediating role of psychological safety and reporting attitudes in the relationship between ethical leadership and internal whistleblowing in the nursing context. The following hypotheses are proposed based on the above theoretical and empirical rationale. H1: Ethical leadership will be positively associated with internal whistleblowing. H2: Ethical leadership will be positively related to psychological safety. H3: Psychological safety will mediate the relationship between ethical leadership and internal whistleblowing. H4: Ethical leadership will be positively associated with reporting attitudes. H5: Reporting attitudes will mediate the relationship between ethical leadership and internal whistleblowing.

2. Methods

2.1. Study setting and participants

This cross-sectional study was conducted at five university hospitals affiliated with Mansoura University, in Mansoura City, Dakahlia Governorate, Egypt. These hospitals are recognized as

tertiary care teaching hospitals and are pivotal in medical and nursing education programs. They provide comprehensive medical care to a diverse patient population in the Dakahlia Governorate, with an estimated population of 7,098,015 individuals. The reporting of this observational study follows the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines [19].

A convenience sample of clinical nurses was actively engaged in patient care and were willing to participate. Inclusion criteria required participants to have permanent employment and to possess at least one year of relevant professional experience. Exclusion criteria encompassed nurse managers, nursing interns, and part-time nurses. The required sample size was calculated online using the "A Priori Sample Size Calculator for Structural Equation Models" [20]. Based on four latent variables, 22 observed variables, an expected effect size of 0.3, a power of 0.95, and a significance level of 0.05, 207 participants were needed. To accommodate a potential 20 % attrition proportion, the final sample was therefore adjusted to 249 nurses.

2.2. Instruments

2.2.1. Demographics questionnaire

This instrument collected data on the participants' age, gender, marital status, education level, and work experience.

2.2.2. Ethical Leadership Scale (ELS)

The 10-item ELS was developed by Brown et al. [21] and used to assess nurses' perceptions of ethical behaviors demonstrated by their supervisors. Each item was rated on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree), yielding a total score range from 5 to 50. Higher scores indicate stronger perceptions of ethical leadership. The original study obtained a Cronbach's α coefficient of 0.92 [21], while the current study yielded a Cronbach's α coefficient of 0.91.

2.2.3. Psychological Safety Scale (PSS)

The 5-item PSS was developed by Liang et al. [22] and used to assess nurses' perceptions of their ability to express their opinions freely without encountering criticism or negative consequences. Each item was rated on a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree), yielding total scores ranging from 5 to 25. Higher scores reflect greater psychological safety. The original study reported a Cronbach's α coefficient of 0.75 [22], while the current study yielded a Cronbach's α coefficient of 0.84.

2.2.4. Positive Attitudes Toward Whistleblowing Scale (PATWS)

The 3-item PATWS was developed by Parak et al. [23] and used to evaluate the positive attitudes held by nurses towards reporting or disclosing information about wrongdoing, unethical behavior, or illegal activities. Items were rated on a 5-point Likert scale (1 = strongly disagree to 5 = strongly agree), with total scores ranging from 3 to 15. Higher scores indicate more positive attitudes toward reporting unethical conduct. The original study reported a Cronbach's α coefficient of 0.80 [23], while the current study yielded a Cronbach's α coefficient of 0.86.

2.2.5. Internal Whistleblowing Scale (IWS)

The 4-item IWS was developed by Park et al. [23] and used to assess nurses' willingness to report wrongdoing within their organization to their immediate supervisors. Items were captured using a 5-point Likert scale, with scores ranging from 4 to 20. Higher scores reflect greater willingness to report. The original Cronbach's α coefficient was 0.78 [23], the current study yielded 0.81.

Permissions using the four scales were granted by the copyright holders. All four scales were translated into Arabic using the standardized cross-cultural adaptation guidelines by Brislin et al. [24]. The process involved independent forward translation by two bilingual experts, synthesis of the translations, followed by blind back-translation by two different translators. An expert committee, comprising seven academic and clinical nursing specialists, reviewed the translated version for semantic, idiomatic, experiential, and conceptual equivalence. Minor revisions were made based on expert consensus to enhance linguistic and cultural appropriateness. The finalized Arabic version was pilot-tested with a convenience sample of 33 nurses to evaluate clarity, readability, and content comprehensibility. Inclusion criteria for the pilot study are the same as those for the main study. Based on feedback, minor formatting adjustments were implemented (e.g., font size); no modifications were made to the scale items. These 33 participants were not included in the main study sample. Reliability and construct validity were further assessed in the main sample using the structural equation model (SEM) approach to evaluate the measurement model.

2.3. Data collection

Data collection was conducted over three months (August to October 2024). Paper-based questionnaires were distributed to nurses across various hospital departments and shifts to ensure broad participation. Each questionnaire was accompanied by an information sheet detailing the study objectives, confidentiality assurances, and the voluntary nature of participation. Participants were informed that completion and return of the questionnaire implied consent. Clear instructions were provided to minimize missing responses, and participants were encouraged to review their answers and contact the research team for clarification. As a result, the final dataset contained no missing data. Completed questionnaires were returned in sealed envelopes and stored in locked containers in the nursing managers' offices. Nurses required approximately 10–15 min to complete the survey.

2.4. Data analysis

Data were analyzed using IBM SPSS 23.0 and AMOS 23.0. The assumption of normality was assessed using histograms, Q-Q plots, and the absolute values of skewness $\leq$2 or kurtosis $\leq$4 [25], confirming that the data met normality assumptions. Measures such as the mean and standard deviation were reported for continuous variables, while categorical variables were summarized using frequencies and percentages. Pearson correlation analysis was conducted to examine the relationships among the study variables. Variance inflation factors (VIFs) and tolerance values were calculated to assess multicollinearity. When VIFs $<$ 2 and tolerance values $>$ 0.70 indicate the absence of multicollinearity.

A two-step approach was used to assess the proposed hypotheses [26]. In step 1, confirmatory factor analysis (CFA) is used to assess the measurement model. Convergent validity was assessed through factor loadings, average variance extracted (AVE) with a threshold of greater than 0.50, and composite reliability (CR) exceeding 0.70 [27]. Discriminant validity was established using multiple criteria. First, the Fornell–Larcker criterion was applied, which requires that the square root of the AVE for each construct is greater than the correlations among constructs, and that the AVE for each construct exceeds the maximum shared variance (MSV) [28]. Second, we employed the heterotrait–monotrait ratio (HTMT) of correlations, with all values

remaining below the conservative threshold of 0.85 [29]. Cronbach's α coefficients were used to assess the internal consistency of each scale, with a benchmark of 0.70 for acceptable reliability [30]. In Step 2, SEM was used to test the hypothesized relationships. Bootstrapping (5,000 resamples) was applied to estimate the indirect effects with 95 % confidence intervals. Model fit was assessed using multiple fit indices, including a chi-square to degrees of freedom ratio (χ^2/df) $\leq$ 3.00; Incremental Fit Index (IFI), Goodness-of-Fit Index (GFI), Comparative Fit Index (CFI), and Tucker–Lewis Index (TLI) values $\geq$ 0.90; and Standardized Root Mean Square Residual (SRMR) and Root Mean Square Error of Approximation (RMSEA) values $\leq$ 0.08 [31]. Harman's single-factor test assessed common method variance (CMV) [32]. The significance level was set at $P < 0.05$ (two-tailed).

2.5. Ethical considerations

This study received approval from the Research Ethics Committee, Nursing Faculty, Mansoura University, Egypt (Serial No. P.0539) and adhered to the Declaration of Helsinki. Informed written consent was obtained from all participants following an explanation of the study's aim. Confidentiality and anonymity were assured, and the participation was entirely voluntary.

3. Results

3.1. Demographic characteristics of the participants

A total of 305 questionnaires were distributed, of which 273 completed responses met the inclusion criteria, yielding a final response rate of 89.5 %. The remaining 32 responses were excluded: 15 nurses declined to participate, 10 provided incomplete responses, and seven respondents were excluded as they did not meet the inclusion criterion of having at least one year of clinical experience.

The study participants were predominantly young, with 78.8 % aged between 20 and 30 years, and a mean age of 28.25 $\pm$ 5.67 years. Females represented 71.8 % of the sample, while males accounted for 27.2 %. Slightly more than half of the participants (54.2 %) were married. Regarding educational attainment, most nurses held a technical diploma (64.5 %), while 35.5 % held a bachelor's degree. Regarding clinical experience, 59.3 % had between 1 and 5 years of professional work experience, with a mean of 6.91 $\pm$ 6.15 years (Table 1).

Table 1
Demographic characteristics of the nurses (n = 273).

Characteristics	n (%)
Age (years)	
20–30	215 (78.8)
31–40	47 (17.2)
>40	11 (4.0)
Gender	
Male	77 (27.2)
Female	196 (71.8)
Marital status	
Unmarried	125 (45.8)
Married	148 (54.2)
Education level	
Technical degree	176 (64.5)
Bachelor's degree	97 (35.5)
Work experience (years)	
1–5	162 (59.3)
6–10	59 (21.6)
>10	52 (19.1)

3.2. Confirmatory factor analysis

The hypothesized four-factor model demonstrated an acceptable and superior fit to the data, as evidenced by the following fit indices: χ^2 = 427.51, df = 180, χ^2/df = 2.38, IFI = 0.92, TLI = 0.91, CFI = 0.92, RMSEA = 0.07, and SRMR = 0.06. Therefore, this study supported the construct validity of the four distinct latent variables (Table 2).

3.3. Descriptive statistics, reliability, validity, and interrelationships among study variables

The average score for ethical leadership was 4.14 $\pm$ 0.57, indicating a generally high perception of ethical leadership in participants' organizations. Psychological safety had an average score of 3.80 $\pm$ 0.64, reflecting a relatively high level of perceived psychological safety. Reporting attitudes averaged 4.12 $\pm$ 0.67, suggesting positive attitudes toward reporting unethical behaviours. Internal whistleblowing had a mean score of 3.93 $\pm$ 0.52, indicating a strong willingness to report unethical practices internally. Before assessing the reliability and validity (measurement model). Harman's single-factor test indicated that a single factor accounted for 33.29 % of the variance, which is below the 50 % threshold, suggesting no significant CMV bias. All constructs demonstrated strong internal consistency, with Cronbach's α values ranging from 0.81 to 0.91, and CR values between 0.81 and 0.92, exceeding the recommended threshold of 0.70. Convergent validity was supported by AVE values ranging from 0.52 to 0.67, all exceeding the acceptable level of 0.50. Item 5 of the PSS was removed due to a factor loading below 0.05 threshold. Finally, factor loadings for all other items ranged from 0.65 to 0.86, which exceeds the acceptable benchmark of 0.50, confirming item-level contributions to their respective constructs.

Discriminant validity was established through multiple criteria. First, the square root of the AVE for each construct (EL = 0.725; PS = 0.763; RA = 0.820; IW = 0.718) exceeded the corresponding inter-construct correlations, thereby meeting the Fornell–Larcker criterion. Second, the AVE values for all constructs were greater than their MSV, providing additional evidence of discriminant validity. Finally, the HTMT of correlations (0.376-0.458) were all below the conservative threshold of 0.85, further confirming that the constructs were empirically distinct. All VIFs (EL = 1.23; PS = 1.21; RA = 1.25) were below 2 and tolerance values (EL = 0.814; PS = 0.829; RA = 0.779) were above 0.70, indicating the absence of multicollinearity concerns. Correlation analysis revealed that ethical leadership was significantly and positively associated with psychological safety (r = 0.370, $P < 0.001$), reporting attitudes (r = 0.431, $P < 0.001$), and internal whistleblowing (r = 0.415, $P < 0.001$). Furthermore, psychological safety and reporting attitudes were positively correlated with internal whistleblowing (r = 0.436, $P < 0.001$, and r = 0.450, $P < 0.001$, respectively). Additionally, psychological safety was linked positively with reporting attitudes (r = 0.429, $P < 0.001$) (Table 3).

3.4. Structural equation model (Hypothesis testing)

The Maximum likelihood estimates demonstrated that ethical leadership had a significant positive direct effect on psychological safety (β = 0.38, $P < 0.001$), reporting attitudes (β = 0.44, $P < 0.001$), and internal whistleblowing (β = 0.21, P = 0.009). Additionally, both psychological safety (β = 0.26, $P < 0.001$) and reporting attitudes (β = 0.27, $P < 0.001$) had a significant positive influence on internal whistleblowing among the studied nurses. The indirect effect of ethics-centered leadership on internal whistleblowing intermediated by psychological safety was also

Table 2
Summary of confirmatory factor analysis.

Model	χ^2	*df*	χ^2/df	IFI	TLI	CFI	RMSEA	SRMR
One factor model (EL + PS + RA + IW)	1447.06	189	7.66	0.59	0.54	0.58	0.16	0.15
Two-factor model (EL + PS + RA; IW)	1133.14	185	6.13	0.69	0.64	0.69	0.14	0.13
Two-factor model (EL; PS + RA + IW)	1004.95	188	5.35	0.73	0.69	0.73	0.13	0.12
Three-factor model (EL; PS + RA; IW)	807.02	186	4.34	0.79	0.78	0.79	0.11	0.09
Four-factor model (EL; PS; RA; IW)	427.51	180	2.38	0.92	0.91	0.92	0.07	0.06

Note: EL = Ethical leadership. PS = Psychological safety. RA = Reporting attitudes. IW = Internal whistleblowing. χ^2: Chi-square. *df*: Degrees of freedom. χ^2/df: Chi-square to degrees of freedom ratio. IFI = Incremental Fit Index. TLI = Tucker–Lewis Index. CFI = Comparative Fit Index. RMSEA = Root Mean Square Error of Approximation. SRMR = Standardized Root Mean Square Residual.

Table 3
Descriptive statistics, reliability, validity, and correlation analysis of EL, PS, RA, and IW.

Variables	*Mean* ± *SD*	Factor loading	Cronbach's α coefficient	CR	AVE	MSV	EL	PS	RA	IW
EL	4.14 ± 0.57	0.65–0.81	0.91	0.92	0.53	0.18	–	–	–	–
PS	3.80 ± 0.64	0.65–0.84	0.84	0.85	0.58	0.19	0.370*	–	–	–
RA	4.12 ± 0.67	0.79–0.86	0.86	0.86	0.67	0.20	0.431*	0.429*	–	–
IW	3.93 ± 0.52	0.69–0.77	0.81	0.81	0.52	0.20	0.415*	0.436*	0.450*	–

Note: EL = Ethical leadership. PS = Psychological safety. RA = Reporting attitudes. IW = Internal whistleblowing. CR = Composite reliability. AVE = Average variance extracted. MSV = Maximum shared variance. *$P < 0.001$.

substantial ($\beta = 0.10$, $P < 0.001$), with a Bias-Corrected (BC) 95 %*CI* of [0.04, 0.15], accounting for 23.26 % of the total effect. Similarly, reporting attitudes demonstrated a significant mediating effect ($\beta = 0.12$, $P = 0.003$), with a BC 95 %*CI* of [0.05, 0.19], accounting for 27.91 % of the total effect. The total effect of ethics-centered leadership on internal whistleblowing was significant ($\beta = 0.43$), with a BC 95 %*CI* of [0.24, 0.52], confirming the strength of the relationship. The parallel model of the study fit indices as follows: $\chi^2 = 441.21$, $df = 184$, $\chi^2/df = 2.40$, IFI = 0.92, TLI = 0.91, CFI = 0.92, RMSEA = 0.07, and SRMR = 0.06. The model variables accounted for 29.30 % of the variance in internal whistleblowing among the studied nurses (Table 4 and Appendix A).

4. Discussion

This study examined the relationships among ethical leadership, psychological safety, reporting attitudes, and internal whistleblowing among nurses. The findings demonstrated that ethical leadership exerted a significant direct positive effect on nurses' internal whistleblowing behaviors. Furthermore, this relationship was partially mediated by psychological safety and positive reporting attitudes, confirming a dual-mediation model in which ethical leadership fosters whistleblowing by creating a psychologically safe work environment and encouraging positive attitudes toward reporting unethical conduct.

The finding that ethical leadership positively influences internal whistleblowing supports H1, indicating a significant positive association between ethical managerial practices and internal whistleblowing. This result highlights the critical role of ethical managerial practices in motivating nurses to report harmful or unethical behaviors, thereby fostering a culture of accountability. This is consistent with social learning theory [33], which posits that leaders serve as ethical role models and, through their behavior, shape the ethical actions of followers. Ethical leaders influence the moral cognition of subordinates, encouraging them to take principled action, including whistleblowing. This interpretation aligns with the findings of Liu et al. [34], who noted that ethical leaders are adept at recognizing ethical dilemmas and acting decisively despite potential risks. Similarly, Çekiç et al. [4] and Dewi et al. [35] reported that ethical leadership significantly promotes nurses' willingness to report wrongdoing. Moreover, Ghanem Atalla et al. [36] found that shared leadership negatively correlated with missed nursing care, implying that fostering accountability and responsibility may promote whistleblowing behavior.

This study also confirmed a positive association between ethical leadership and psychological safety, supporting H2. This outcome highlights the key role of ethical leaders in creating an environment of trust, respect, and open communication, consistent with Leader-Member Exchange (LMX) theory [37,38]. Ethical leaders foster psychological safety by demonstrating fairness, accountability, and moral decision-making, enabling nurses to feel secure in voicing concerns without fear of negative repercussions. This finding is consistent with Edmondson's conceptualization of

Table 4
Maximum likelihood estimates of the mediation model.

Path	β	*SE*	*t*	*P*	BC 95 %*CI*	Proportion of total effect (%)
Direct effect						
EL → PS	0.38	0.08	5.50	<0.001	0.29–0.65	–
EL → RA	0.44	0.07	6.19	<0.001	0.31–0.68	–
EL → IW	0.21	0.07	2.61	0.009	0.05–0.32	48.8
PS → IW	0.26	0.05	3.54	<0.001	0.09–0.31	–
RA → IW	0.27	0.06	3.50	<0.001	0.08–0.37	–
Indirect effect						
EL → PS → IW	0.10	–	–	<0.001	0.04–0.15	23.26
EL → RA → IW	0.12	–	–	0.003	0.05–0.19	27.91
Total effect	0.43	–	–	–	0.24–0.52	100.00

Note: EL = Ethical leadership. PS = Psychological safety. RA = Reporting attitudes. IW = Internal whistleblowing. BC = Bias-Corrected.

psychological safety as a foundational condition for speaking up in organizations [13]. Ethical leaders foster an ethical environment in the workplace that empowers staff members to speak out against unethical and illegal workplace behavior [39]. It also supports prior work by El-Gazar et al. [12], who identified ethical leadership as a key predictor of psychological safety in nursing environments.

Moreover, the study demonstrated that psychological safety significantly mediated the relationship between ethical leadership and nurses' whistleblowing, supporting H3. This suggests that ethical managerial practices indirectly promote whistleblowing behaviors by creating a psychologically safe environment. This finding aligned with the findings by El-Gazar et al. [12] and those of Potipiroon and Wongpreedee [10], who found that ethical leadership enhances psychological safety and thereby increases the likelihood of whistleblowing among Egyptian nurses and public sector employees in Thailand. It is also consistent with Tehranineshat et al. [40], who emphasized the broader role of ethical climate and professional values in shaping nurses' professional quality of life.

Ethical leadership was also found to have a significant positive effect on reporting attitudes, supporting H4. This finding underscores the role of ethics-centered leadership in shaping positive norms and perception regarding error reporting, consistent with social learning theory [33]. Ethical leaders shape norms that frame whistleblowing as an ethically appropriate behavior through modeling and reinforcement. Similarly, Zakiy and Satyarini [41] found that the absence of ethical leadership serves as a barrier to whistleblowing, suggesting that its presence could significantly enhance employees' attitudes toward reporting unethical behavior. Similarly, Jiang [42] emphasized that the teaching competency in the 360-Degree Leadership Model equips ethical leaders to educate their staff and positively shape attitudes toward error reporting and organizational transparency.

The study further demonstrated that reporting attitudes mediated the relationship between ethical leadership and whistleblowing, confirming H5. This suggests that ethical leadership indirectly facilitates whistleblowing by cultivating positive attitudes toward reporting. This mediation effect resonates with the theory of planned behavior, which asserts that perceived behavioral control, enhanced through attitudes, facilitates desired actions such as whistleblowing [14]. Our research supports the belief that an ethical leader can impact an individual's ethical self-concept, including their moral identity, and stimulate a specific facet of their ethical character. This, in turn, influences future actions that are advantageous to both corporations and their employees (whistleblowing) [1,2]. This finding is consistent with Hechanova and Manaois [43], they noted that ethical leadership shapes organizational norms and controls, influencing employees' attitudes toward corruption and their intention to whistleblowing. Moreover, Shonhadji [44] identified a significant positive relationship between employees' attitudes and their whistleblowing intentions, underscoring the pivotal role of attitudinal dispositions in shaping the likelihood of engaging in whistleblowing behavior.

The findings have important implications for healthcare organizations. First, organizations should prioritize ethical leadership training to equip leaders with skills to model ethical behavior and resolve conflicts fairly. Second, fostering psychological safety is essential to encourage nurses to report misconduct without fear of retaliation. This can be achieved by creating a culture of trust and open communication. Third, healthcare organizations should promote a positive reporting attitude toward whistleblowing through education and awareness programs, emphasizing their role in enhancing patient safety and organizational integrity.

This study has several limitations. First, a cross-sectional design limits the ability to establish a causal relationship. Future research should employ longitudinal designs to explore these relationships over time. Second, the study focused exclusively on nurses, limiting the generalizability of the findings to other healthcare professionals. Future studies should include physicians, allied health personnel, and administrative staff to gain a more holistic view of whistleblowing dynamics in healthcare. Third, demographic characteristics were not included in the structural model, as the primary aim was to test a theoretically driven model focused on the dual mediating role of psychological safety and reporting attitudes in the relationship between ethical leadership and internal whistleblowing. Future research should consider these control variables to enhance model robustness and control for potential confounding effects. Finally, while the study examined psychological safety and reporting attitudes as mediators, other factors, such as organizational justice and trust, may also play significant roles and should be examined in future studies.

5. Conclusions

This study underscores the pivotal role of ethical leadership in promoting internal whistleblowing among nurses. By fostering psychological safety and positive reporting attitudes, ethical leaders create a work climate conducive to ethical voice and organizational transparency. The dual-mediation model enriches our understanding of the mechanisms through which leadership behaviors translate into whistleblowing intentions. Healthcare institutions must take proactive steps to cultivate ethical leadership and create a culture of transparency and accountability.

Data availability statement

The datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

CRediT authorship contribution statement

Ibrahim Abdullatif Ibrahim: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review & editing, Project administration. **Indrajit Doddanavar:** Methodology, Validation, Formal analysis, Visualization, Data curation. **Samah Mohamed Abdelrahim:** Conceptualization, Methodology, Investigation, Resources, Data curation, Writing - review & editing. **Mennat Allah G. Abou Zeid:** Conceptualization, Methodology, Validation, Resources, Writing - review & editing.

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Declaration of competing interest

The authors of this study have declared no conflict of interest.

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Appendices. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijnss.2025.08.004.

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